Efficient Inference for Inverse Reinforcement Learning and Dynamic Discrete Choice Models

Lars van der Laan*
University of Washington

Aurélien Bibaut
Netflix Research

Nathan Kallus
Cornell Tech, Cornell University

April 28, 2026

Abstract

Inverse reinforcement learning (IRL) and dynamic discrete choice (DDC) models explain sequential decision-making through latent reward functions. Existing flexible IRL methods allow rich reward representations but typically do not support valid inference, whereas classical DDC approaches provide inference only under restrictive parametric structure. We develop a semiparametric framework for *debiased inverse reinforcement learning* in maximum-entropy IRL and Gumbel-shock DDC models. Our key identifying observation is that the log-behaviour policy can be treated as a pseudo-reward. It point-identifies policy value differences and, under a normalization constraint, the reward itself. This lets us express reward-dependent estimands as smooth functionals of the behaviour policy and transition kernel. We establish pathwise differentiability, derive efficient influence functions, and construct automatic debiased machine-learning estimators that permit flexible nuisance estimation while attaining $\sqrt{n}$-consistency, asymptotic normality, and semiparametric efficiency. The result is a computationally tractable framework connecting modern IRL with classical DDC inference.

Keywords— inverse reinforcement learning, dynamic discrete choice, semiparametric inference, debiased machine learning, policy evaluation

*Corresponding author: lvdlaan@uw.edu

1 Introduction

Behavioural data arise in robotics, human–computer interaction, healthcare, and economics. Inverse reinforcement learning (IRL) seeks to recover the latent reward or utility function underlying observed sequential decisions. It has been used to model expert driving and robotic control (e.g. Abbeel and Ng, 2004; Wulfmeier et al., 2016), user behaviour in interactive systems (e.g. Chandramohan et al., 2011; Hossain et al., 2023), and clinical decision-making in critical care (e.g. Yu et al., 2019; Bovenzi et al., 2025). Because rewards summarize preferences or tradeoffs, they provide a basis for counterfactual analysis—for example, predicting how behaviour would change under new policies, constraints, or environments—that is not available from policy prediction alone.

A central difficulty is that IRL is inherently ill posed: many rewards can rationalize the same behavior (Ng et al., 1999; Ng and Russell, 2000; Cao et al., 2021; Skalse et al., 2023; Skalse and Abate, 2024). To make the problem tractable, IRL and dynamic discrete choice (DDC) models impose an optimality model linking rewards to behavior. Exact optimality implies deterministic policies and is often too rigid for observed human behavior. A foundational stochastic alternative in DDC adds Type-I extreme-value utility shocks, yielding choice probabilities of softmax form and a corresponding soft Bellman equation (Rust, 1987; Hotz and Miller, 1993; Aguirregabiria and Mira, 2010). The same softmax structure arises in modern maximum-entropy IRL by adding an entropy term to the agent's objective, so actions with higher long-run utility are chosen with higher probability rather than deterministically (Ziebart et al., 2008; Ziebart, 2010; Haarnoja et al., 2017). Even under this structure, rewards remain identified only up to an equivalence class. Point identification therefore requires a normalization (Rust, 1987; Geng et al., 2020), such as setting the reward of a reference action to zero.

Uncertainty quantification for reward-dependent summaries is central to IRL applications. These summaries include normalized reward averages, structural coefficients, and values of counterfactual decision rules (Kalouptsidi et al., 2021; Zielnicki et al., 2025). The problem is challenging because rewards are latent, identified only after normalization, and estimated through flexible nuisance components such as choice probabilities and transition laws. Existing methods largely split in two: classical DDC methods provide inference under restrictive parametric utility models (e.g. Rust, 1987; Hotz and Miller, 1993; Aguirregabiria and Mira, 2010), whereas flexible machine-learning approaches to IRL generally focus on reward estimation without valid uncertainty quantification for downstream functionals (e.g. Abbeel and Ng, 2004; Ramachandran and Amir, 2007; Ziebart et al., 2008; Levine et al., 2011; Ho and Ermon, 2016; Fu et al., 2018; Wulfmeier et al., 2016; Geng et al., 2020). We close this gap by developing a semiparametric framework for *debiased inverse reinforcement learning*.

1.1 Our contributions

We study reward identification and inference in maximum-entropy IRL and Gumbel-shock dynamic discrete choice models. Our main contributions are as follows:

1. We show that, in softmax IRL, the reward is recoverable from the log-behavior policy under an appropriate normalization. More generally, a broad class of reward-dependent estimands can be expressed as smooth functionals of the log-behavior policy and transition dynamics, reducing reward-based inference to inference on observable choice probabilities and the transition kernel.

2. We establish pathwise differentiability and derive efficient influence functions, thereby characterizing the semiparametric efficiency bounds for these targets. Under general affine reward normalizations, we show that the efficient influence function can be

obtained directly from the unnormalized case, yielding a general template for inference under normalization.

3. We construct automatic debiased machine-learning estimators (Chernozhukov et al., 2022; van der Laan et al., 2025a) that allow flexible nuisance estimation while achieving $\sqrt{n}$-consistency, asymptotic normality, and semiparametric efficiency.

4. We specialize the general theory to derive efficiency bounds and efficient estimators for several key IRL/DDC targets, including fixed-policy values, counterfactual softmax policy values, normalized-reward policy values, and structural coefficients of a normalized reward.

The paper proceeds as follows. Section 3 shows how policy-value differences, and under normalization the reward itself, can be identified from observed behavior. Sections 4.3 and 5 develop the semiparametric theory and automatic debiased machine-learning estimators for a broad class of smooth target functionals. Section 6 specializes these results to the main policy-value targets.

1.2 Related work

Dynamic discrete choice models. Classical DDC work studies identification and efficient estimation of finite-dimensional utility parameters using nested fixed-point, conditional choice probability, and simulation-based methods (Rust, 1987; Hotz and Miller, 1993; Hotz et al., 1994; Aguirregabiria and Mira, 2010). In contrast, we fix the softmax/Gumbel-shock structure of Rust (1987) and target reward-dependent functionals under nonparametric reward representations. To our knowledge, debiased estimation, efficient influence functions, and efficiency bounds for such functionals have not been developed in this literature. For identification, the closest connection is the conditional choice probability approach of Hotz and Miller (1993): under Gumbel shocks, log-odds of observed choices identify differences in choice-specific value functions. Rather than using log-odds for value-difference identification, we use the log behavior policy together with the transition law to identify normalized rewards and reward-dependent functionals. An expanded related work is provided in Appendix D.

Inverse reinforcement learning. Classical IRL recovers parametric or linear rewards under hard optimality assumptions (e.g. Ng and Russell, 2000; Abbeel and Ng, 2004), while more recent work uses maximum-entropy or adversarial formulations with flexible function approximators (e.g. Ziebart et al., 2008, 2010; Levine et al., 2011; Ho and Ermon, 2016; Fu et al., 2018; Geng et al., 2020). Several papers formalize the resulting reward ambiguity, including Fu et al. (2018), Theorem 1 of Cao et al. (2021), and Lemma 2 of Geng et al. (2020). Deep PQR (Geng et al., 2020) identifies normalized rewards under anchor-action constraints using a construction tailored to that setting: the reward is represented through the log behavior policy, an anchor-specific Bellman fixed point, and a second-stage conditional-mean regression. In contrast, our identification results express normalized rewards in terms of standard primitive objects—the log behavior policy, transition law, and policy-specific Q-functions—and apply to a broader class of affine normalization constraints, with anchor-action constraints as a special case. These works focus on reward recovery; our goal is inference for reward-dependent functionals. functionals.

Off-policy evaluation and double reinforcement learning. Debiased and doubly robust methods for off-policy evaluation estimate policy values when the reward is known (e.g. Jiang and Li, 2016; Thomas and Brunskill, 2016; Tang et al., 2019; Liu et al., 2018; van Der Laan et al., 2018; Kallus and Uehara, 2020; Uehara et al., 2020; Shi et al., 2021; Kallus and Uehara, 2022; Mehrabi and Wager, 2024). Double Reinforcement Learning extends this perspective to more general functionals of the Q-function (Kallus and Uehara, 2020, 2022; van der Laan et al., 2025b). Our framework is complementary: it treats the reward as unknown, recovers it through the softmax behaviour model, and then performs semiparametric inference for the induced reward-dependent targets.

2 Preliminaries

2.1 Problem setup

We consider an infinite-horizon, time-homogeneous Markov decision process (MDP) with state space $\mathcal{S}$, action space $\mathcal{A}$, and reference measure μ on $\mathcal{S}$. The initial state $S_0 \in \mathcal{S}$ is drawn from density ρ_0 with respect to μ. At each time $t \geq 0$, the agent occupies state S_t, selects action $A_t \in \mathcal{A}$ according to the behaviour policy $\pi_0(\cdot \mid S_t)$, receives latent reward $r_0^\dagger(A_t, S_t)$, and transitions to a next state $S_{t+1} \in \mathcal{S}$ with conditional density $k_0(\cdot \mid A_t, S_t)$ with respect to μ. In the IRL setting, the analyst observes states and actions, but not rewards. We write P_0 for the law of the one-step observation (S_0, A_0, S_1), and $\mathbb{P}_0$ for the induced law of the full trajectory.

The inferential task is to learn reward-dependent summaries, such as policy values, from observed state–action transitions. We observe $\mathcal{D}_n := \{(S_i, A_i, S_i')\}_{i=1}^n$, a collection of n i.i.d. samples from P_0. Our analysis also accommodates temporal dependence through Markov-chain arguments under mixing conditions (Bibaut et al., 2021; Mehrabi and Wager, 2024). A fundamental challenge is that $r_0^\dagger$ is not point identified from state–action data alone: many distinct rewards induce the same observable behaviour (Ng et al., 1999; Ng and Russell, 2000). IRL methods address this by imposing an optimality model linking the behaviour policy to the reward.

2.2 Soft optimality and the soft Bellman equation

We adopt the notion of *soft optimality*, which arises in Gumbel–shock-to-reward formulations (Rust, 1987; Hotz and Miller, 1993; Aguirregabiria and Mira, 2010) and in maximum-entropy IRL (Ziebart et al., 2008, 2010; Haarnoja et al., 2017). Soft optimality replaces deterministic reward maximization with an entropy-regularized objective, yielding stochastic softmax decision rules. We first recall the classical (hard) value and Q-functions. For any policy $\pi : \mathcal{A} \times \mathcal{S} \to [0,1]$ and transition kernel $k : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$, let $\mathbb{E}_{k,\pi}$ denote expectation with respect to trajectories generated by the MDP with transition kernel k, policy π, and initial state density ρ_0.

For any function $f : \mathcal{A} \times \mathcal{S} \to \mathbb{R}$, define the forward operators

$$\mathcal{P}_k f(a,s) := \int f(a,s')\, k(s' \mid a,s)\, \mu(ds'), \qquad \pi f(s) := \sum_{a \in \mathcal{A}} \pi(a \mid s)\, f(a,s).$$

For a discount factor $\gamma \in [0,1)$ and reward function $r : \mathcal{A} \times \mathcal{S} \to \mathbb{R}$, the *(hard) Q-function* is

$$q_{r,k}^{\pi,\gamma}(a,s) = \mathbb{E}_{k,\pi}\left[\sum_{t=0}^{\infty} \gamma^t r(A_t, S_t) \,\middle|\, A_0 = a,\, S_0 = s \right],$$

the unique bounded solution to Bellman's fixed-point equation

$$q_{r,k}^{\pi,\gamma}(a,s) = r(a,s) + \gamma\, \mathcal{P}_k V_{r,k}^{\pi,\gamma}(a,s), \qquad V_{r,k}^{\pi,\gamma}(s) = \pi q_{r,k}^{\pi,\gamma}(s), \tag{1}$$

where $V_{r,k}^{\pi,\gamma}$ is the associated *(hard) value function* (Bellman, 1954). Equivalently, it solves the inverse problem

$$\mathcal{T}_{k,\pi,\gamma}(q_{r,k}^{\pi,\gamma}) = r, \qquad \mathcal{T}_{k,\pi,\gamma}(f) := f - \gamma\, \mathcal{P}_k \pi f,$$

where $\mathcal{T}_{k,\pi,\gamma}$ is the Bellman operator. Since $\gamma < 1$, the operator $\mathcal{T}_{k_0,\pi,\gamma}$ is invertible on the Banach space of state–action functions equipped with the supremum norm (Bellman, 1954), and we denote its inverse by $\mathcal{T}_{k,\pi,\gamma}^{-1}$. Throughout, when the discount factor is γ, we omit explicit dependence on γ in our notation.

Entropy regularization augments the reward $r_0(A_t, S_t)$ with an entropy bonus $\tau\,\mathcal{H}(\pi(\cdot \mid S_t))$, where $\mathcal{H}(\pi(\cdot \mid s)) := -\sum_a \pi(a \mid s) \log \pi(a \mid s)$ is the Shannon entropy of the policy and $\tau \in (0, \infty)$ is a regularization parameter. The value and Q-functions associated with this augmented reward are referred to as the *soft* value and Q-functions (Haarnoja et al., 2017). Since rewards are identified only up to scale, we set $\tau = 1$ without loss of generality, interpreting rewards on a relative scale.

An agent is *soft optimal* (with discount factor γ and $\tau = 1$) if its policy π_0 maximizes the *soft value*, that is, the entropy-regularized expected return,

$$\pi \;\mapsto\; \mathbb{E}_{k_0,\pi}\left[\sum_{t=0}^{\infty} \gamma^t \{\, r_0^\dagger(A_t, S_t) + \mathcal{H}(\pi(\cdot \mid S_t))\,\}\right],$$

over all policies π (Haarnoja et al., 2017). The resulting soft-optimal policy is the softmax (multinomial logit) distribution induced by the soft Q-function $r_0^\dagger(a,s) + \gamma\, v_0^\dagger(a,s)$:

$$\pi_0(a \mid s) = \frac{\exp\Big\{r_0^\dagger(a,s) + \gamma\, v_0^\dagger(a,s)\Big\}}{\sum_{a' \in \mathcal{A}} \exp\Big\{r_0^\dagger(a',s) + \gamma\, v_0^\dagger(a',s)\Big\}}, \tag{2}$$

where $v_0^\dagger$ satisfies the soft Bellman fixed-point system

$$v_0^\dagger(a,s) = \int V_0^\dagger(s')\, k_0(s' \mid a, s)\, \mu(ds'), \quad V_0^\dagger(s) = \log\left(\sum_{a' \in \mathcal{A}} \exp\{r_0^\dagger(a',s) + \gamma\, v_0^\dagger(a',s)\}\right),$$

with $V_0^\dagger$ the soft value function of π_0 and $v_0^\dagger$ its associated continuation value. Equivalently, in the maximum-entropy IRL formulation (Ziebart et al., 2008, 2010), the agent's policy maximizes the entropy of the trajectory distribution subject to matching the observed behaviour.[1] We take the discount factor γ as known. Identification of γ from observed

[1]This yields a Boltzmann trajectory distribution, $\mathbb{P}_0^{\pi_0}((S_0, A_0, \ldots) = (s_0, a_0, \ldots)) \propto$

choices—requiring additional variation beyond that in our setting—is analysed in Abbring and Daljord (2020).

Soft optimality also arises in dynamic discrete-choice (DDC) models (Rust, 1987; Hotz and Miller, 1993; Aguirregabiria and Mira, 2010), where the agent selects actions by maximizing a random utility $A_t \in \arg\max_{a\in\mathcal{A}} \left\{ r_0^\dagger(a, S_t) + \gamma\, v_0^\dagger(a, S_t) + \varepsilon_t(a) \right\}$, where, in the DDC model, $v_0^\dagger(a, s)$ coincides with the classical choice-specific continuation value and $\varepsilon_t(a)$ are i.i.d. Gumbel (Type I extreme value) shocks (Rust, 1987; Pitombeira-Neto et al., 2024).

2.3 Notation

We introduce the notation and regularity conditions used throughout. Let $\lambda := \# \otimes \mu$ denote the product measure on $\mathcal{A} \times \mathcal{S}$, where $\#$ is the uniform measure on the finite action set $\mathcal{A}$ and μ is a finite dominating measure on $\mathcal{S}$. Thus, for any measurable $B \subseteq \mathcal{A} \times \mathcal{S}$, $\lambda(B) = \frac{1}{|\mathcal{A}|} \sum_{a\in\mathcal{A}} \int 1_B(a, s)\, \mu(ds)$. Let $\mathcal{H} := L^\infty(\lambda)$ denote the space of essentially bounded reward functions, and let $\mathcal{K} \subset L^\infty(\mu \otimes \lambda)$ denote the space of essentially bounded transition kernels k (conditional densities of $S' \mid (A, S)$ with respect to μ) satisfying $c < k(s' \mid a, s) < C$ for λ-a.e. (a, s). We equip $\mathcal{H}$ and $\mathcal{K}$ with the L^2 norms

$$\|h\|_{\mathcal{H}} := \left(\int \sum_{a\in\mathcal{A}} h(a, s)^2\, \mu(ds) \right)^{1/2}, \qquad \|k\|_{\mathcal{K}} := \left(\int \sum_{a\in\mathcal{A}} k(s' \mid a, s)^2\, \mu(ds')\, \lambda(a, ds) \right)^{1/2}.$$

For readability, we keep only the most frequently recurring conventions here. We write $r_0^\dagger$ for the latent reward and $r_0(a, s) := \log \pi_0(a \mid s)$ for the pseudo-reward. We use q and V for policy-specific Q- and state-value functions; when a scalar policy value is needed, we write $V_{r,k}(\pi; \gamma)$.

We make the simplifying assumption that the initial state distribution, behaviour policy, and transition dynamics are uniformly bounded away from zero and infinity.

$\exp\Big(\sum_{t=0}^{\infty} \gamma^t r_0^\dagger(a_t, s_t)\Big)$.

(A1) *(Bounded positivity).* There exist constants $c, C \in (0, \infty)$ and $\eta \in (0,1)$ such that, for

$$\mu \otimes \lambda\text{-a.e. } (s, a, s') \in \mathcal{S} \times \mathcal{A} \times \mathcal{S}, c < \rho_0(s) < C, \qquad \eta < \pi_0(a \mid s), c < k_0(s' \mid a, s) < C.$$

All results are stated under Assumption A1. Let $\mathcal{M}$ denote the collection of distributions $P(ds, da, ds') = \rho_P(s)\pi_P(a \mid s)k_P(s' \mid a, s)\mu(ds')\lambda(da, ds)$ satisfying this assumption. The boundedness requirement is imposed for technical convenience and can in principle be relaxed using suitable moment conditions on ρ_0, π_0, and k_0.

3 Identification

We first characterize reward indeterminacy under soft optimality, then show that policy value differences remain identified, and finally impose a normalization to identify the reward itself.

3.1 Partial identification of the reward

Without additional constraints, the reward $r_0^\dagger$ is only partially identified: many rewards induce the same softmax policy for a soft-optimal agent (Cao et al., 2021). Its scale is also unidentified; by convention, we fix the entropy-regularization parameter at $\tau = 1$, which is equivalent to fixing the softmax temperature (Haarnoja et al., 2017). This subsection formalizes the resulting partial identification.

Formally, the partial identification of $r_0^\dagger$ can be characterized through the softmax likelihood (Rust, 1987; Ziebart et al., 2008). The observed policy π_0 maximizes this soft Bellman–constrained likelihood, so any reward r that induces the same softmax policy is also a solution of the population M-estimation problem

$$r \in \operatorname*{argmin}_{r' \in \mathcal{H}} E_0 \left[\log \left(\sum_{a \in \mathcal{A}} \exp\{r'(a, S) + \gamma v_{r',k_0}(a, S)\} \right) - \{r'(A, S) + \gamma v_{r',k_0}(A, S)\} \right], \quad (3)$$

where v_{r',k_0} satisfies the soft Bellman equation

$$v_{r',k_0}(a,s) = \int \log\left(\sum_{a' \in \mathcal{A}} \exp\{r'(a',s') + \gamma v_{r',k_0}(a',s')\} \right) k_0(s' \mid a,s)\, \mu(ds'). \tag{4}$$

Any reward function that induces the same behaviour as $r_0^\dagger$ has the potential-based reward–shaping form (Ng et al., 1999; Fu et al., 2018; Geng et al., 2020)

$$r(a,s) = r_0^\dagger(a,s) + c(s) - \gamma \int c(s')\, k_0(s' \mid a,s)\, \mu(ds'), \tag{5}$$

for a unique state-dependent potential $c : \mathcal{S} \to \mathbb{R}$. Under this transformation, the soft Q-functions satisfy $q^{\text{soft}}_{r,k_0}(a,s) = q^{\text{soft}}_{r_0^\dagger,k_0}(a,s) + c(s)$.

We begin with a trivial but informative solution to (3): the log-behaviour policy $r_0(a,s) := \log \pi_0(a \mid s)$. The soft Bellman constraint (4) is satisfied with $v_{r_0,k_0} \equiv 0$, because the normalization $\sum_{a' \in \mathcal{A}} \pi_0(a' \mid s) = 1$ implies that the log-partition term vanishes, $\log \sum_{a' \in \mathcal{A}} \exp\{r_0(a',s)\} = 0$.

Lemma 1 (Trivial reward solution)**.** *The log-behaviour policy $r_0 = \log \pi_0$ is a solution to (3) and satisfies (5) with $c(s) = -V_0^\dagger(s)$.*

3.2 Identification of policy values

Although the reward is only partially identified, many quantities of practical interest remain point identified. Lemma 1 shows that behaviour determines $r_0^\dagger$ only up to potential-based shaping. Such shifts change all policy values by the same constant and thus do not affect policy comparisons. Consequently, policy value differences remain identified, and the log-behaviour-policy reward r_0 suffices for policy evaluation. The next result formalizes this fact.

Theorem 2 (Behaviour policy identifies value differences). *Suppose that $r_0^\dagger$ solves (3) for a discount factor $\gamma \in [0,1)$. Then, for any policy π,*

$$q^{\pi,\gamma}_{r_0,k_0}(a,s) = q^{\pi,\gamma}_{r_0^\dagger,k_0}(a,s) - V_0^\dagger(s), \qquad V_{r_0,k_0}(\pi;\gamma) = V_{r_0^\dagger,k_0}(\pi;\gamma) - E_0[V_0^\dagger(S)].$$

Hence, for any two policies π_1 and π_2, $V_{r_0,k_0}(\pi_1;\gamma) - V_{r_0,k_0}(\pi_2;\gamma) = V_{r_0^\dagger,k_0}(\pi_1;\gamma) - V_{r_0^\dagger,k_0}(\pi_2;\gamma)$.

A subtle but important point is that the identification of policy value differences holds only for the discount factor γ that appears in the soft Bellman equation governing the agent's behaviour. To evaluate policies under a different discount factor γ', the reward must typically be point identified. Section 3.3 shows how this can be achieved via a normalization constraint. Before turning to that result, we illustrate Theorem 2 in a counterfactual softmax setting.

Example 1 (Identification for counterfactual softmax policies). *Let $\mathcal{A}^\star \subseteq \mathcal{A}$ and $\tau^\star \in (0,\infty)$. For $(r,k) \in \mathcal{H} \times \mathcal{K}$, define the counterfactual softmax policy*

$$\pi^\star_{r,k}(a \mid s) \propto \mathbb{1}\{a \in \mathcal{A}^\star\} \exp\big\{(r(a,s) + \gamma v^\star_{r,k}(a,s))/\tau^\star\big\}, \tag{6}$$

where $v^\star_{r,k}$ solves the counterfactual soft Bellman equation

$$\begin{aligned} v^\star_{r,k}(a,s) \; &= \int \Phi^\star_{r,k}(s')\, k(s' \mid a,s)\, \mu(ds'), \\ \Phi^\star_{r,k}(s) \; &:= \tau^\star \log\left(\sum_{a' \in \mathcal{A}^\star} \exp\big(\tfrac{1}{\tau^\star}(r(a',s) + \gamma v^\star_{r,k}(a',s))\big) \right). \end{aligned} \tag{7}$$

Because $\pi^\star_{r,k}$ is invariant to potential-based reward shaping of the form (5), and Lemma 1 shows that r_0 and $r_0^\dagger$ differ by such a transformation, we have $\pi^\star_{r_0^\dagger,k_0} = \pi^\star_{r_0,k_0}$. Applying Theorem 2 to the pair of policies $\pi_1 = \pi^\star_{r_0,k_0}$ and $\pi_2 = \pi_0$, the value difference

$V_{r_0^\dagger,k_0}(\pi^\star_{r_0^\dagger,k_0};\gamma) - V_{r_0^\dagger,k_0}(\pi_0;\gamma)$ *is identified using the log-behaviour-policy reward* r_0 *in place of* $r_0^\dagger$. $\square$

3.3 Exact reward identification under normalization

The next theorem shows that, after normalization, the latent reward $r_0^\dagger$ is exactly recoverable from the log-behaviour policy reward r_0. We fix a reference policy ν and a known state-only anchor $g : \mathcal{S} \to \mathbb{R}$, and impose that the average reward in state s under ν equals $g(s)$:

$$\nu r_0^\dagger(s) := \sum_{\tilde{a}\in\mathcal{A}} \nu(\tilde{a} \mid s) r_0^\dagger(\tilde{a}, s) = g(s), \quad P_0\text{-a.e. } s. \tag{8}$$

This constraint anchors the free state potential in (5) and covers common normalizations, including anchor-action, statewise sum-to-zero, and value-based constraints (Hotz and Miller, 1993; Bajari et al., 2010; Kallus and Udell, 2016; Geng et al., 2020). For example, requiring $V^{\nu,\gamma}_{r_0^\dagger,k}(s) = h(s)$ for all s is equivalent, by the Bellman equation, to choosing $g(s) = h(s) - \gamma \sum_{a\in\mathcal{A}} \nu(a \mid s) \int h(\tilde{s})\, k(d\tilde{s} \mid a, s)$. The pair (ν, g) can also be chosen using auxiliary information, such as outcomes that determine a reference reward level (Kallus et al., 2018). For instance, with auxiliary state–reward data from the agent, one could take $\nu = \pi$ and $g(s) = E_0\{r(A, S) \mid S = s\}$.

The following identification result shows that the normalized reward can be expressed through the Q-function $q^{\nu,\gamma}_{r_0-g,k_0} = \mathcal{T}^{-1}_{k_0,\nu,\gamma}(r_0 - g)$ of the normalization policy ν.

Theorem 3 (Normalization identifies the reward itself)**.** *Let* ν *be a policy. There exists a unique solution* $r_{0,\nu,\gamma,g} \in \mathcal{H}$ *to* (3) *that satisfies* $\nu r_{0,\nu,\gamma,g}(s) = g(s)$ *for* P_0*-a.e.* $s \in \mathcal{S}$*.*

Moreover,

$$r_{0,\nu,\gamma,g}(a,s) = g(s) + q^{\nu,\gamma}_{r_0-g,k_0}(a,s) - \sum_{\tilde{a}\in\mathcal{A}} \nu(\tilde{a}\mid s)\, q^{\nu,\gamma}_{r_0-g,k_0}(\tilde{a},s),$$

$$v_{0,\nu,\gamma,g}(a,s) = \tfrac{1}{\gamma}\left(r_0(a,s) - g(s) - q^{\nu,\gamma}_{r_0-g,k_0}(a,s)\right),$$

where $v_{0,\nu,\gamma,g}$ solves the soft Bellman equation with reward $r_{0,\nu,\gamma,g}$.

Thus, if $r_0^\dagger$ satisfies (8), it is identified by $r_{0,\nu,\gamma,g}$. Inference under normalization therefore reduces to estimating the observable pseudo-reward $r_0 = \log \pi_0$ and the Bellman quantity $q^{\nu,\gamma}_{r_0-g,k_0}$, which motivates the next section.

4 Debiased Inverse Reinforcement Learning

We now move from identification to inference. We first define a generic target functional built from the pseudo-reward and transition kernel, then derive its efficient influence function and bias expansion, and finally specialize these objects to the motivating examples below.

4.1 Statistical objective

The identification results in Section 3 imply that the log-behaviour policy $r_0 = \log \pi_0$ can be used as a *pseudo-reward* for many IRL targets. Policy value differences are identified directly from r_0 by Theorem 2, and Theorem 3 shows that normalized reward functionals can also be recovered from (r_0, k_0). We therefore study smooth real-valued functionals of the form

$$\psi_0 := E_0[m(A,S,r_0,k_0)],$$

where $(r,k) \mapsto m(\cdot, r, k)$ is a smooth map from $\mathcal{H}\times\mathcal{K}$ to $L^2(\lambda)$. Formally, $\psi_0 = \Psi(P_0)$ corresponds to the parameter $\Psi : \mathcal{M} \to \mathbb{R}$ defined for $P \in \mathcal{M}$ by $\Psi(P) := E_P[m(A,S,r_P,k_P)]$, where $k_P(s' \mid a, s) := \frac{dP_{S'|A,S}}{d\mu}(s' \mid a, s)$ is the state transition kernel, and $r_P(a, s) = \log P(A = a \mid S = s)$.

4.2 Motivating examples

Our framework accommodates a wide range of counterfactual targets in IRL. We record the main targets used in the sequel: three policy-value parameters and one structural normalized-reward parameter.

Example 1a (Policy value)**.** *The policy value* $\psi_0 := E_0[V^{\pi,\gamma}_{r_0,k_0}(S)]$ *of a policy* π *with discount factor* γ *corresponds to the map* $m(a, s, r, k) = V^{\pi,\gamma}_{r,k}(s)$. □

The same framework also covers interventions on the agent's behaviour or environment.

Example 1b (Value of a soft-optimal agent)**.** *Consider the counterfactual softmax policy* $\pi^\star_{r_0,k_0}$ *defined in* (6)*, obtained by optimizing the pseudo-reward* r_0 *under kernel* k_0 *with action set* $\mathcal{A}^\star$ *and entropy parameter* $\tau^\star$*. Unlike Example 1a, this policy depends on both* r_0 *and* k_0*. The policy value* ψ_0 *fits our framework through the map* $m(a, s, r, k) := V^\star_{r,k}(s)$*, where* $V^\star_{r,k}$ *denotes the value function of* $\pi^\star_{r,k}$*.* □

A closely related counterfactual keeps the pseudo-reward target fixed but replaces the transition kernel by a known $k^\star$, as in simulated or redesigned environments (e.g. Abbeel and Ng, 2004; Ziebart et al., 2008; Wulfmeier et al., 2016; Fu et al., 2018).

Example 1c (Policy values in counterfactual environments)**.** *The value of the soft-optimal agent that maximizes the pseudo-reward* r_0 *in an environment with known transition kernel* $k^\star$ *corresponds to* $m(a, s, r, k) := V^\star_{r,k^\star}(s)$*, where* $V^\star_{r,k^\star}$ *denotes the value function of* $\pi^\star_{r,k^\star}$*.* □

Under a normalization constraint, functionals of the true reward $r_0^\dagger$ also fall within our framework. In these settings, the target is identified by the ν-normalized reward $\bar{r}_0 := r_{0,\nu,\gamma,g}$.

Example 2 (Policy values under reward normalization)**.** *Suppose we are interested in* $\psi_0 = E_0[\tilde{m}(A, S, \bar{r}_0, k_0)]$*, a feature of the* ν*-normalized reward* $\bar{r}_0$*. By Theorem 3, this can be expressed as* $\psi_0 = E_0[m(A, S, r_0, k_0)]$*, where*

$$m(a, s, r, k) := \tilde{m}(a, s, r_{r,k,\nu,\gamma,g}, k), \qquad r_{r,k,\nu,\gamma,g}(a, s) := g(s) + q^{\nu,\gamma}_{r-g,k}(a, s) - V^{\nu,\gamma}_{r-g,k}(s),$$

and $r_{r,k,\nu,\gamma,g} = g + (I - \nu)\, \mathcal{T}^{-1}_{k,\nu,\gamma}(r - g)$ *denotes the* ν*-normalized reward with anchor* g*. This includes fixed-policy values or softmax values built with a target discount* γ' *that may differ from the normalization discount* γ*.* $\square$

Example 3 (Structural coefficients under reward normalization)**.** *Let* $x : \mathcal{A} \times \mathcal{S} \to \mathbb{R}^p$ *be a known bounded feature map and write* $\bar{r}_0 := r_{0,\nu,\gamma,g}$*. If* $G_0 := E_0\{x(A, S)x(A, S)^\top\}$ *is nonsingular, the coefficient in the linear projection of the normalized reward onto* x *is*

$$\theta_0 := G_0^{-1} E_0\{x(A, S)\bar{r}_0(A, S)\}.$$

For example, in Rust's bus replacement model, a coordinate of $x(a, s)$ *may indicate replacement, in which case the corresponding coordinate of* θ_0 *is a replacement-cost coefficient.* $\square$

4.3 Statistical efficiency and functional bias expansion

A key step in our methodology is to establish the pathwise differentiability of Ψ and derive its efficient influence function (EIF). In semiparametric statistics, the EIF determines the generalized Cramér–Rao lower bound for Ψ and forms the basis for constructing debiased, efficient estimators (Bickel et al., 1993; Laan and Robins, 2003; Van der Laan et al., 2011; Chernozhukov et al., 2018).

The next conditions ensure that $E_0[m(A, S, r, k)]$ is first-order expandable in r and k. The *reward tangent space* $T_{\mathcal{H}}$ is defined as the completion of $\mathcal{H}$ under the inner product

$\langle h_1, h_2\rangle_{\mathcal{H},0} := E_0[h_1(A,S)h_2(A,S)]$. We define the *kernel tangent space at* k by

$$T_{\mathcal{K}}(k) := \left\{\beta : \int \beta(s',a,s)^2\, k(s' \mid a,s)\,\mu(ds') < \infty \text{ and } \int \beta(s',a,s)\, k(s' \mid a,s)\,\mu(ds') = 0 \text{ for } \lambda\text{-a.e. } (a,s)\right\},$$

and set $T_{\mathcal{K}} := T_{\mathcal{K}}(k_0)$, equipped with the inner product $\langle \beta_1, \beta_2\rangle_{\mathcal{K},0} := E_0[\beta_1(S',A,S)\beta_2(S',A,S)]$. Intuitively, $T_{\mathcal{K}}$ consists of allowable infinitesimal kernel scores that preserve the normalization of conditional densities under $k(\cdot \mid a,s)\mu(ds')$.

(C1) (*Functional differentiability*) There exist Gâteaux derivatives $\partial_r m(\cdot, r_0, k_0) : T_{\mathcal{H}} \to L^2(\lambda)$ and $\partial_k m(\cdot, r_0, k_0) : T_{\mathcal{K}} \to L^2(\lambda)$ at (r_0, k_0) such that, for all $\alpha \in T_{\mathcal{H}}$ and $\beta \in T_{\mathcal{K}}$,

$$\frac{d}{dt} E_0\big[m\big(A, S,\, r_0 + t\alpha,\, (1+t\beta)k_0\big)\big]\Big|_{t=0} = E_0[\partial_r m(A,S,r_0,k_0)(\alpha) + \partial_k m(A,S,r_0,k_0)(\beta)]\,.$$

(C2) (*Continuity of derivatives*) The linear functionals $E_0[\partial_r m(A,S,r_0,k_0)(\cdot)]$ and $E_0[\partial_k m(A,S,r_0,k_0)(\cdot)]$ are continuous, in the sense that

$$\sup_{\alpha\in T_{\mathcal{H}}} \frac{E_0[\partial_r m(A,S,r_0,k_0)(\alpha)]}{\|\alpha\|_{\mathcal{H},0}} < \infty, \qquad \sup_{\beta\in T_{\mathcal{K}}} \frac{E_0[\partial_k m(A,S,r_0,k_0)(\beta)]}{\|\beta\|_{\mathcal{K},0}} < \infty.$$

(C3) (*Lipschitz continuity*) The map $(r,k) \mapsto E_0[m(A,S,r,k)]$ is locally Lipschitz continuous at (r_0,k_0) with respect to $(\mathcal{H}, \|\cdot\|_{\mathcal{H},0}) \times (\mathcal{K}, \|\cdot\|_{\mathcal{K}})$.

Conditions C1-C3 hold if the map $(r,k) \mapsto E_0[m(A,S,r,k)]$ is Fréchet differentiable at (r_0,k_0) with respect to the product norm $\|h\|_{\mathcal{H},0} + \|g\|_{\mathcal{K}}$; see Lemma 6 in Appendix F. For additive Fréchet derivatives in the kernel argument, we write $\widetilde{T}_{\mathcal{K}}$ for the ambient normed linear space of additive kernel perturbations, equipped with $\|\cdot\|_{\mathcal{K}}$, and denote the corresponding additive derivative by $\widetilde{\partial}_k$. Thus ∂_k is reserved for score derivatives on $T_{\mathcal{K}}$, while $\widetilde{\partial}_k$ acts on additive directions $g \in \widetilde{T}_{\mathcal{K}}$.

The EIF depends on two additional nuisance functions: the Riesz representers of the partial functional derivatives of $(r, k) \mapsto E_0[m(A, S, r, k)]$. Let $\alpha_0 \in T_{\mathcal{H}}$ and $\beta_0 \in T_{\mathcal{K}}$ denote these representers with respect to $\langle \cdot, \cdot \rangle_{\mathcal{H},0}$ and $\langle \cdot, \cdot \rangle_{\mathcal{K},0}$, so that

$$E_0[\partial_r m(A, S, r_0, k_0)(\alpha)] = \langle \alpha_0, \alpha \rangle_{\mathcal{H},0}, \tag{9}$$

$$E_0[\partial_k m(A, S, r_0, k_0)(\beta)] = \langle \beta_0, \beta \rangle_{\mathcal{K},0}. \tag{10}$$

Theorem 4 (Pathwise differentiability)**.** *Under C1-C3, the parameter $\Psi : \mathcal{M} \to \mathbb{R}$ is pathwise differentiable at P_0. Its efficient influence function is*

$$\chi_0(s, a, s') := \alpha_0(a, s) - \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} \mid s)\alpha_0(\tilde{a}, s) + \beta_0(s, a, s') + m(a, s, r_0, k_0) - \Psi(P_0),$$

where α_0 and β_0 are the Riesz representers defined in (9)–(10).

The EIF χ_0 determines the generalized Cramér–Rao lower bound for Ψ. Its kernel component follows from existing EIF results for offline reinforcement learning. Specifically, if the rewards $\{r_0(A_i, S_i)\}_{i=1}^n$ were observed, then $\beta_0(S_i, A_i, S_i') + m(A_i, S_i, r_0, k_0) - \Psi(P_0)$ would be the nonparametric EIF for the known-reward target $P \mapsto E_P\{m(A, S, r_0, k_P)\}$. Thus, in each IRL application, it remains only to derive the reward-side Riesz representer α_0 for the map $r \mapsto E_0\{\partial_r m(A, S, r, k_0)\}$. For example, EIFs for policy values are given in van Der Laan et al. (2018); Kallus and Uehara (2020, 2022), and van der Laan et al. (2025b) derives EIFs for general continuous linear functionals of the Q-function. Although those works allow additive reward noise, the EIF for continuous linear functionals of the Q-function is unchanged because the reward-noise score is orthogonal to the target-relevant tangent space (Bickel et al., 1993; Laan and Robins, 2003).

The EIF characterizes the first-order behaviour of plug-in estimators of ψ_0. To formalize this, we introduce the score

$$\varphi_{r,k}(s', a, s; \alpha, \beta) := \alpha(a, s) - \sum_{\tilde{a} \in \mathcal{A}} \exp\{r(\tilde{a}, s)\}\, \alpha(\tilde{a}, s) + \beta(s', a, s),$$

which provides the linear approximation of $E_0[m(A, S, r, k)]$ around (r_0, k_0). To analyze the bias of this approximation, define the remainder

$$\mathrm{Rem}_0(r_0, k_0;\, r, k) := E_0[m(A, S, r, k)] - E_0[m(A, S, r_0, k_0)] - E_0[\varphi_{r_0,k_0}(\cdot; \alpha_0, \beta_0)].$$

The next theorem establishes a von Mises expansion for this decomposition (Mises, 1947).

Theorem 5 (von Mises expansion)**.** *Assume C1. Let $r \in \mathcal{H}$, $k \in \mathcal{K}$, $\alpha \in T_{\mathcal{H}}$, and $\beta \in T_{\mathcal{K}}(k)$, and suppose $\|r\|_{L^\infty(\lambda)} + \|r_0\|_{L^\infty(\lambda)} < M$ for some $M \in (0, \infty)$. Then*

$$\begin{aligned} E_0\big[m(A, S, r, k)\big] - E_0\big[m(A, S, r_0, k_0)\big] \;+\; \int \varphi_{r,k}(s', a, s; \alpha, \beta)\, P_0(ds, da, ds') \\ = -\,\langle r - r_0,\, \alpha - \alpha_0 \rangle_{\mathcal{H},0} \;+\; O\big(\|r - r_0\|_{\mathcal{H}}^2\big) \\ +\; O\Big(\|\beta_0 - \beta\|_{L^2(P_0)} \Big\|\tfrac{k - k_0}{k_0}\Big\|_{L^2(P_0)}\Big) \;+\; \mathrm{Rem}_0(r_0, k_0;\, r, k), \end{aligned}$$

where the implicit constants depend only on M.

It is often possible to show that the linearization remainder is second order, for example $\mathrm{Rem}_0(r_0, k_0;\, r, k) \lesssim \|k - k_0\|_{\mathcal{K}}^2 + \|r - r_0\|_{\mathcal{H}}^2$. For Example 1a, this follows from Lemma 10 in Appendix H.

4.4 Functionals of ν-normalized reward

Suppose we have already derived the EIF of a functional of the pseudo-reward r_0 using Theorem 4, and we now seek the EIF of the corresponding functional of the ν-normalized reward $r_{0,\nu,\gamma,g}$ from Theorem 3. The next result shows that the EIF can be obtained directly from the unnormalized case via a simple transformation of the associated Riesz representers.

Consider the functional $\psi_0 = E_0[\tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0)]$ from Example 2. Let $\tilde{\alpha}_0$ and $\tilde{\beta}_0$ be the Riesz representers of the partial derivatives of the map $(r, k) \mapsto E_0[\tilde{m}(A, S, r, k)]$ at $(r_{0,\nu,\gamma,g}, k_0)$, defined by the property that for all $\alpha \in T_{\mathcal{H}}$ and $\beta \in T_{\mathcal{K}}$,

$$E_0[\partial_r \tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0)[\alpha]] = \langle \tilde{\alpha}_0, \alpha \rangle_{\mathcal{H},0},$$

$$E_0[\partial_k \tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0)[\beta]] = \langle \tilde{\beta}_0, \beta \rangle_{\mathcal{K},0}.$$

The Riesz representers are defined analogously to (9) and (10), the only difference being that the partial derivatives are now evaluated at $(r_{0,\nu,\gamma,g}, k_0)$ rather than (r_0, k_0). The following theorem assumes Conditions C1*–C3*, which adapt Conditions C1–C3 to the map $\tilde{m}$ evaluated at $(r_{0,\nu,\gamma,g}, k_0)$. Since these conditions are direct analogues, they are stated in Appendix A.1.

Theorem 6 (Pathwise differentiability for normalized rewards)**.** *Suppose C1*-C3*, and define* $m(a, s, r, k) := \tilde{m}(a, s, r_{r,k,\nu,\gamma,g}, k)$*. Then the parameter* $\Psi : P \mapsto E_P[m(A, S, r_P, k_P)]$ *is pathwise differentiable with EIF* χ_0*, given in Theorem 4, where*

$$\alpha_0(a, s) := \tilde{\alpha}_0(a, s) - \frac{\nu(a \mid s)}{\pi_0(a \mid s)} \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} \mid s)\, \tilde{\alpha}_0(\tilde{a}, s);$$

$$\beta_0(s' \mid a, s) := \tilde{\beta}_0(s' \mid a, s) + \alpha_0(a, s) \left\{ r_0(a, s) - g(s) + \gamma V^{\nu,\gamma}_{r_0 - g, k_0}(s') - q^{\nu,\gamma}_{r_0 - g, k_0}(a, s) \right\}.$$

Equivalently, Theorem 6 gives a conditional influence operator for the normalized reward itself. For any downstream functional whose derivative with respect to the normalized reward has Riesz representer $w \in T_{\mathcal{H}}$, define

$$\mathcal{A}_{\nu}w(a,s) := w(a,s) - \frac{\nu(a \mid s)}{\pi_0(a \mid s)} \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} \mid s) w(\tilde{a}, s)$$

and

$$\Delta_{\nu}(s', a, s) := r_0(a,s) - g(s) + \gamma V^{\nu,\gamma}_{r_0 - g, k_0}(s') - q^{\nu,\gamma}_{r_0 - g, k_0}(a, s).$$

Then the contribution to the EIF coming only from estimating $r_{0,\nu,\gamma,g}$ is

$$\mathcal{I}_{\nu}(w)(s', a, s) := \mathcal{A}_{\nu}w(a,s) - \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} \mid s) \mathcal{A}_{\nu}w(\tilde{a}, s) + \mathcal{A}_{\nu}w(a,s)\Delta_{\nu}(s', a, s).$$

Thus, to compute the EIF for any smooth functional $E_0\{\tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0)\}$, one may first compute the ordinary Riesz representers $(w, \tilde{\beta})$ of $\tilde{m}$ as if the normalized reward were observed, and then use

$$\chi_0 = \mathcal{I}_{\nu}(w) + \tilde{\beta}_0(S', A, S) + \tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0) - \psi_0.$$

This separates the reusable normalized-reward influence calculation from the functional-specific derivative w.

The simple structure of the EIF follows from the following representation.

Lemma 7 (Representation of reward averages under normalization)**.** *For any $w \in L^2(P_0)$,*

$$E_0[w(A,S) r_{0,\nu,\gamma,g}(A,S)] = E_0\left[w(A,S) \left\{ g(S) + r_0(A,S) - \sum_{a \in \mathcal{A}} \nu(a \mid S) r_0(a, S) \right\} \right].$$

Hence, weighted averages of $r_{0,\nu,\gamma,g}$, with possibly signed weights w, reduce to the reward-free term $E_0\{w(A,S)g(S)\}$ and an average of r_0 after subtracting its ν-mean at the same state. Consequently, linear summaries such as the best linear predictor of $r_{0,\nu,\gamma,g}(A,S)$ can be identified from the pseudo-reward.

5 Automatic debiased estimation and inference

5.1 General form of the estimator

Let r_n and k_n denote estimators of the pseudo-reward r_0 (the log-behaviour policy) and the transition kernel k_0, respectively. The pseudo-reward r_0 can be learned from state–action data $\{(S_i, A_i)\}_{i=1}^n$ via probabilistic classification, while k_0 may be estimated using conditional density methods. A natural estimator of ψ_0 is the plug-in estimator $n^{-1}\sum_{i=1}^n m(A_i, S_i, r_n, k_n)$. Under flexible nuisance estimation, this plug-in approach can exhibit substantial bias. We therefore use a one-step influence-function-based correction.

Let $\alpha_n \in \mathcal{H}$ and $\beta_n \in L^2(P_0)$ estimate the representers α_0 and β_0 from Section 4.3. We take β_n to be centered with respect to k_n, so that $\int \beta_n(s', a, s)\, k_n(s' \mid a, s)\mu(ds') = 0$; this can always be enforced post hoc. We propose the one-step estimator $\psi_n := P_n\{m(\cdot, r_n, k_n) + \varphi_{r_n,k_n}(\cdot\,; \alpha_n, \beta_n)\}$, or, equivalently,

$$\psi_n = \frac{1}{n}\sum_{i=1}^n m(A_i, S_i, r_n, k_n) + \frac{1}{n}\sum_{i=1}^n \left[\alpha_n(A_i, S_i) - \sum_{a\in\mathcal{A}} \exp\{r_n(a, S_i)\}\alpha_n(a, S_i) + \beta_n(S_i', A_i, S_i)\right]. \tag{11}$$

The second term applies the EIF from Theorem 4 to correct the first-order bias of the plug-in estimator (Bickel et al., 1993). For policy value parameters, the kernel representer β_0 coincides with the usual transition-kernel component of the EIF in off-policy evaluation, and can be estimated using existing DRL methods. The reward representer can often be derived analytically or estimated directly by *Riesz regression* (Chernozhukov et al., 2022; van der Laan et al., 2025a), using the population risk

$$\alpha \mapsto E_0\left[\alpha(A, S)^2 - 2\,\partial_r m(A, S, r_0, k_0)(\alpha)\right].$$

Because (11) is agnostic to the choice of m, we refer to this procedure as an *automatic* DML estimator.

When ψ_0 depends on value- or Q-functions induced by k_0, these higher-level nuisances can often be estimated directly, avoiding explicit estimation of k_0. The next section uses this reparameterization for the main policy-value examples.

5.2 Asymptotic theory

The next theorem establishes that the proposed estimator ψ_n is asymptotically linear with influence function χ_0 and is therefore efficient for ψ_0. We impose the following conditions.

(C4) (*Sample splitting and boundedness*) The estimators r_n, k_n, α_n, and β_n are obtained from data independent of $\mathcal{D}_n$. Moreover, $\left\|m(\cdot, r_n, k_n) + \varphi_{r_n,k_n}(\cdot\,; \alpha_n, \beta_n)\right\|_{L^2(P_0)} = O_P(1)$ and $\left\|m(\cdot, r_0, k_0) + \varphi_{r_0,k_0}(\cdot\,; \alpha_0, \beta_0)\right\|_{L^2(P_0)} = O(1)$.

(C5) (*Nuisance consistency and estimation rates*)

(a) $\left\|m(\cdot, r_n, k_n) + \varphi_{r_n,k_n}(\cdot\,; \alpha_n, \beta_n) - m(\cdot, r_0, k_0) - \varphi_{r_0,k_0}(\cdot\,; \alpha_0, \beta_0)\right\|_{L^2(P_0)} = o_P(1)$

(b) $\|r_n - r_0\|_{\mathcal{H}} = o_P(n^{-1/4})$ and $\|r_n - r_0\|_{\mathcal{H},r_0} \|\alpha_n - \alpha_0\|_{\mathcal{H},r_0} = o_P(n^{-1/2})$.

(c) $\left\|\frac{k_n - k_0}{k_0}\right\|_{L^2(P_0)} \|\beta_n - \beta_0\|_{L^2(P_0)} = o_P(n^{-1/2})$ and $\mathrm{Rem}_0(r_0, k_0;\, r_n, k_n) = o_P(n^{-1/2})$.

Theorem 8 (Asymptotic linearity and efficiency)**.** *Suppose Conditions C4–C5 hold. Then the autoDML estimator ψ_n defined in (11) satisfies*

$$\psi_n - \psi_0 = \frac{1}{n}\sum_{i=1}^{n} \chi_0(S_i, A_i, S_i') + o_P(n^{-1/2}).$$

Consequently, $\sqrt{n}\,(\psi_n - \psi_0) \rightsquigarrow N(0, \sigma_0^2)$, where $\sigma_0^2 = \mathrm{Var}_{P_0}\big(\chi_0(S, A, S')\big)$. Thus ψ_n is nonparametric efficient for ψ_0.

The conditions above are standard in debiased machine learning. Condition C4 enforces independence between nuisance estimation and evaluation; this can be relaxed to cross-fitting, where the sample is partitioned and nuisance functions are estimated on held-out folds (Van der Laan et al., 2011; Chernozhukov et al., 2018). Condition C4 guarantees uniform L^2 boundedness of the true and estimated uncentered influence functions. Condition (a) requires consistency of this estimated uncentered influence function, typically ensured when the nuisance estimators are consistent. Condition C5 specifies the convergence rates needed for the von Mises remainder to be second order. In particular, part (c) yields a form of partial double robustness: slower convergence of α_n can be offset by faster convergence of r_n. In our examples, the linearization remainder often satisfies $\mathrm{Rem}_0(r_0, k_0; r, k) = O\big(\|r - r_0\|_{\mathcal{H}}^2 + \|k - k_0\|_{\mathcal{K}}^2\big)$.

6 Applications

This section specializes the generic theory to the three policy-value targets developed in the main text: the value of a fixed policy, the value of a fixed policy under reward normalization, and the value of a counterfactual soft-optimal policy. The corresponding normalized softmax and structural coefficient examples are collected in Appendix A.

6.1 Efficient influence functions for example parameters

We begin with Example 1a, which concerns the value of a fixed policy π. Our analysis relies on an overlap condition for the stationary state–action distribution, ensuring that the operator $\mathcal{T}_{k_0,\pi}$ has a bounded inverse on $L^2(\lambda)$. For a given policy π, let $p_k^{\pi,\gamma}$ denote a stationary state-μ-density for the MDP with policy π and kernel k. That is, for every $f \in L^\infty(\mu)$, $\int f(s')\, p_k^{\pi,\gamma}(s')\, \mu(ds') = \iint f(s')\, k'_\pi(s' \mid s)\, \mu(ds')\, p_k^{\pi,\gamma}(s)\, \mu(ds)$, where $k'_\pi(s' \mid s) := \sum_{a\in\mathcal{A}} \pi(a \mid s)\, k(s' \mid a, s)$.

(D1) (*Stationary overlap for* π) There exist constants $b, B \in (0, \infty)$ such that $b \leq \frac{p_{k_0}^{\pi,\gamma}}{\rho_0}(s) \cdot \frac{\pi(a|s)}{\pi_0(a|s)} \leq B$ for λ-a.e. (a, s).

In what follows, denote the (unnormalized) discounted state–action occupancy density ratio by

$$d_0^{\pi,\gamma}(a, s) := \rho_0^{\pi,\gamma}(s) \, \frac{\pi(a \mid s)}{\pi_0(a \mid s)}, \qquad \rho_0^{\pi,\gamma}(s) := \frac{1}{\rho_0(s)} \sum_{t=0}^{\infty} \gamma^t \, \frac{d\mathbb{P}_{k_0,\pi}}{d\mu}(S_t = s), \tag{12}$$

where $\rho_0^{\pi,\gamma}$ is the associated state occupancy ratio. This state–action occupancy ratio characterizes the policy value through the identity $E_0[V_{r_0,k_0}^{\pi,\gamma}(S)] = E_0[d_0^{\pi,\gamma}(A, S) \, r_0(A, S)]$. In this sense, it plays the same role in offline policy evaluation as the inverse propensity score does in causal inference.

Theorem 9 (EIF for known policy)**.** *Let* $m(a, s, r, k) := V_{r,k}^{\pi,\gamma}(s)$*. Under D1, the parameter* $\Psi : P \mapsto E_P[V_{r_P,k_P}^{\pi,\gamma}(S)]$ *is pathwise differentiable at* P_0 *with efficient influence function*

$$\begin{aligned}
\chi_0(s, a, s') = \rho_0^{\pi,\gamma}(s) \left(\frac{\pi(a \mid s)}{\pi_0(a \mid s)} - 1 \right) \\
+ d_0^{\pi,\gamma}(a, s) \Big\{ r_0(a, s) + \gamma \, V_{r_0,k_0}^{\pi,\gamma}(s') - q_{r_0,k_0}^{\pi,\gamma}(a, s) \Big\} \\
+ V_{r_0,k_0}^{\pi,\gamma}(s) - \Psi(P_0).
\end{aligned}$$

In Theorem 9, the representers from Theorem 4 take the explicit forms $\alpha_0(a, s) = \rho_0^{\pi,\gamma}(s) \, \frac{\pi(a|s)}{\pi_0(a|s)}$ and $\beta_0(s, a, s') = d_0^{\pi,\gamma}(a, s) \{ r_0(a, s) + \gamma \, V_{r_0,k_0}^{\pi,\gamma}(s') - q_{r_0,k_0}^{\pi,\gamma}(a, s) \}$. Thus the EIF combines a density-ratio term from the pseudo-reward with the familiar temporal-difference term from off-policy evaluation (van Der Laan et al., 2018; Kallus and Uehara, 2020, 2022; van der Laan et al., 2025b).

We next consider Example 1b, concerning the value of the counterfactual softmax policy

$\pi^\star_{r_0,k_0}$. Here $V^\star$ denotes the policy value, $v^\star$ the continuation value, and $\Phi^\star$ the log-sum-exp state value, where

$$\Phi^\star_{r_0,k_0}(s) := \tau^\star \log\left(\sum_{\tilde a\in\mathcal{A}^\star} \exp\left(\tau^{\star-1}\{r_0(\tilde a, s) + \gamma\, v^\star_{r_0,k_0}(\tilde a, s)\}\right)\right).$$

We also define the conditional discounted state-occupancy ratio

$$\rho^\star_{r_0,k_0}(s' \mid a, s) := \frac{1}{\rho_0(s')} \sum_{t=0}^{\infty} \gamma^t \, \frac{\mathbb{P}_{k_0,\pi^\star_{r_0,k_0}}(S_t \in ds' \mid A_0 = a, S_0 = s)}{\mu(ds')},$$

with marginal $\rho^\star_{r_0,k_0}(s) := E_0[\rho^\star_{r_0,k_0}(s \mid A, S)]$ and state-action ratio $d^\star_{r_0,k_0}(a,s) := \pi^\star_{r_0,k_0}(a \mid s)\pi_0(a \mid s)^{-1}\rho^\star_{r_0,k_0}(s)$. We also define the advantage-weighted state occupancy ratio

$$\widetilde{\rho}^\star_{r_0,k_0}(s) := E_0\left[d^\star_{r_0,k_0}(A,S)\{q^\star_{r_0,k_0}(A,S) - V^\star_{r_0,k_0}(S)\}\, \rho^\star_{r_0,k_0}(s \mid A, S)\right], \tag{13}$$

with state–action version $\widetilde{d}^\star_{r_0,k_0}(a,s) := \frac{\pi^\star_{r_0,k_0}(a|s)}{\pi_0(a|s)}\, \widetilde{\rho}^\star_{r_0,k_0}(s)$. In the following condition, let $p^\star_{k_0,r_0} := p_{k_0}^{\pi^\star_{r_0,k_0},\gamma}$ denote the stationary μ-density under $\pi^\star_{r_0,k_0}$.

(D2) (*Stationary overlap for* $\pi^\star_{r_0,k_0}$) There exist $b, B \in (0,\infty)$ such that $b \le \frac{p^\star_{k_0,r_0}(s)}{\rho_0(s)} \cdot \frac{\pi^\star_{r_0,k_0}(a|s)}{\pi_0(a|s)} \le B$ for λ-a.e. (a,s).

(D3) (*Lipschitz continuity of the soft Bellman fixed point*) There exists $L \in (0,\infty)$ such that, for all $k \in \mathcal{K}$, $\|v^\star_{r_0,k} - v^\star_{r_0,k_0}\|_{L^2(\lambda)} \le L\|k - k_0\|_{\mathcal{K}}$.

Condition D3 ensures pathwise differentiability of Ψ with respect to the kernel by controlling von Mises remainder terms of the form $\|v^\star_{r_0,k} - v^\star_{r_0,k_0}\|_{L^2(\lambda)}$ in terms of $\|k - k_0\|_{\mathcal{K}}$.

Theorem 10 (EIF for softmax policy)**.** *Let* $m(a,s,r,k) := V^\star_{r,k}(s)$. *Under D2 and D3, the parameter* $\Psi : P \mapsto E_P[V^\star_{r_P,k_P}(S)]$ *is pathwise differentiable at* P_0 *with efficient influence* *function*

$$\chi_0(s,a,s') = \left\{\frac{1}{\tau^\star}\,\widetilde{\rho}^\star_{r_0,k_0}(s) + \rho^\star_{r_0,k_0}(s)\right\}\left\{\frac{\pi^\star_{r_0,k_0}(a \mid s)}{\pi_0(a \mid s)} - 1\right\}$$
$$+ \frac{\gamma}{\tau^\star}\,\widetilde{d}^\star_{r_0,k_0}(a,s)\left\{\Phi^\star_{r_0,k_0}(s') - v^\star_{r_0,k_0}(a,s)\right\}$$
$$+ d^\star_{r_0,k_0}(a,s)\left\{\, r_0(a,s) + \gamma\, V^\star_{r_0,k_0}(s') - q^\star_{r_0,k_0}(a,s)\,\right\}$$
$$+ V^\star_{r_0,k_0}(s) - \Psi(P_0)$$

Relative to Theorem 9, this EIF adds the advantage-weighted ratio $\widetilde{\rho}^\star_{r_0,k_0}$ and the Bellman term $\frac{\gamma}{\tau^\star}\,\widetilde{d}^\star_{r_0,k_0}(a,s)\{\Phi^\star_{r_0,k_0}(s') - v^\star_{r_0,k_0}(a,s)\}$. Both terms arise because the target policy depends on the soft Bellman solution $v^\star_{r_0,k_0}$.

We now turn to Example 2. Write $d^{\pi,\gamma'}_{0,k_0}(a,s) := \rho_0^{\pi,\gamma'}(s)\,\frac{\pi(a|s)}{\pi_0(a|s)}$ for the discounted state–action occupancy ratio of π at discount γ', where $\rho_0^{\pi,\gamma'}$ is the associated state occupancy ratio. Define $r_{0,\nu,\gamma,g}(a,s) := g(s) + q^{\nu,\gamma}_{r_0-g,k_0}(a,s) - V^{\nu,\gamma}_{r_0-g,k_0}(s)$, and abbreviate $q^{\pi,\gamma'}_{r_0,k_0,\nu} := q^{\pi,\gamma'}_{r_{0,\nu,\gamma,g},k_0}$ and $V^{\pi,\gamma'}_{r_0,k_0,\nu}(s) := \sum_{a\in\mathcal{A}} \pi(a \mid s)\, q^{\pi,\gamma'}_{r_0,k_0,\nu}(a,s)$. In addition to the stationary overlap condition from Condition D1, applied here with target discount γ', we require stationary overlap for the normalization policy ν:

(D4) (*Stationary overlap for ν*) There exist constants $b, B \in (0,\infty)$ such that $b \le \frac{p^{\nu,\gamma}_{k_0}(s)}{\rho_0(s)} \cdot \frac{\nu(a|s)}{\pi_0(a|s)} \le B$ for λ-a.e. (a,s).

Theorem 11 (EIF for policy value under reward normalization)**.** *Let $m(a,s,r,k) := \sum_{\tilde a\in\mathcal{A}} \pi(\tilde a \mid s)\, q^{\pi,\gamma'}_{r,k,\nu}(\tilde a,s)$, where $q^{\pi,\gamma'}_{r,k,\nu}$ is the Q-function of π at discount γ' under the ν-normalized reward $r_{r,k,\nu,\gamma,g}$. Under D1, D4, and C1*-C3*, the parameter $\Psi : P \mapsto$*

$E_P[m(A, S, r_P, k_P)]$ *is pathwise differentiable at* P_0 *with efficient influence function*

$$
\begin{aligned}
\chi_0(s, a, s') = \rho_0^{\pi,\gamma'}(s) &\left(\frac{\pi(a \mid s)}{\pi_0(a \mid s)} - \frac{\nu(a \mid s)}{\pi_0(a \mid s)}\right) \\
&+ d_{0,k_0}^{\pi,\gamma'}(a, s)\left(1 - \frac{\nu(a \mid s)}{\pi(a \mid s)}\right)\left\{r_0(a, s) - g(s) + \gamma V_{r_0-g,k_0}^{\nu,\gamma}(s') - q_{r_0-g,k_0}^{\nu,\gamma}(a, s)\right\} \\
&+ d_{0,k_0}^{\pi,\gamma'}(a, s)\left\{r_{0,\nu,\gamma,g}(a, s) + \gamma' V_{r_0,k_0,\nu}^{\pi,\gamma'}(s') - q_{r_0,k_0,\nu}^{\pi,\gamma'}(a, s)\right\} \\
&+ V_{r_0,k_0,\nu}^{\pi,\gamma'}(s) - \Psi(P_0).
\end{aligned}
$$

Theorem 11 retains the same two-part structure as Theorem 9, but adds a second Bellman term for the normalization map. The first augmentation accounts for estimating the normalized reward $r_{0,\nu,\gamma,g}$ through the auxiliary value $q_{r_0-g,k_0}^{\nu,\gamma}$; the second is the usual off-policy correction for evaluating π under that normalized reward.

Results for the additional examples from Section 4.2, namely softmax values under normalization and structural coefficients, are collected in Appendix A.

6.2 Estimators and asymptotic theory

We next give example-specific DML estimators that avoid direct estimation of the transition kernel k_0. Instead, they target higher-level nuisance functions that yield doubly robust von Mises expansions and simpler rate conditions under which Theorem 8 applies. We focus on Examples 1a, 1b and 2; additional estimators, conditions, and asymptotic results are deferred to Appendices B and B.

We revisit Example 1a. Let $\pi_n := \exp\{r_n\}$, let $q_n^{\pi,\gamma}$ estimate $q_{r_0,k_0}^{\pi,\gamma}$, and define $V_n^{\pi,\gamma}(s) := \sum_{\tilde{a}\in\mathcal{A}} \pi(\tilde{a} \mid s) q_n^{\pi,\gamma}(\tilde{a}, s)$. Let $\rho_n^{\pi,\gamma}$ estimate the state occupancy ratio $\rho_0^{\pi,\gamma}$. Our DML estimator of $\psi_0 = E_0[V^{\pi,\gamma}_{r_0,k_0}(S)]$ is

$$\psi_n := \frac{1}{n}\sum_{i=1}^n V_n^{\pi,\gamma}(S_i) + \frac{1}{n}\sum_{i=1}^n \rho_n^{\pi,\gamma}(S_i)\frac{\pi(A_i \mid S_i)}{\pi_n(A_i \mid S_i)}\left\{r_n(A_i,S_i) + \gamma V_n^{\pi,\gamma}(S_i') - q_n^{\pi,\gamma}(A_i,S_i)\right\}$$
$$+ \frac{1}{n}\sum_{i=1}^n \rho_n^{\pi,\gamma}(S_i)\left\{1 - \frac{\pi(A_i \mid S_i)}{\pi_n(A_i \mid S_i)}\right\}.$$

The nuisance $q^{\pi,\gamma}_{r_0,k_0}$ can be estimated by fitted Q-evaluation, i.e., iterative regression on Bellman targets (Ernst et al., 2005; Munos, 2005; Riedmiller, 2005; Munos and Szepesvári, 2008; Mnih et al., 2013; van der Laan and Kallus, 2025a). The ratio $\rho_0^{\pi,\gamma}$ can be estimated by DICE-style, moment-matching, or saddle-point methods (Liu et al., 2018; Uehara et al., 2020; Nachum et al., 2019; Zhang et al., 2020; Wang et al., 2023; van der Laan et al., 2025b). For example, $\rho_0^{\pi,\gamma} = \mathcal{T}_{k_0,\pi}(\widetilde{\alpha}_0)$ (van der Laan et al., 2025b), where

$$\widetilde{\alpha}_0 := \operatorname*{argmin}_{\alpha \in L^2(\mu)} E_0\left[\{\mathcal{T}_{k_0,\pi}(\alpha)(A_0,S_0)\}^2 - 2\alpha(S_0)\right].$$

Thus one may estimate $\rho_0^{\pi,\gamma}$ by replacing $\mathcal{T}_{k_0,\pi}$ with $\mathcal{T}_{k_n,\pi}$, or by using an equivalent min–max formulation (Liu et al., 2018; Uehara et al., 2020; Dikkala et al., 2020; Kallus and Uehara, 2020). The next theorem gives rate conditions under which ψ_n is asymptotically linear and efficient.

(E1) (*Nuisance rates*) $\left\{\|\rho_n^{\pi,\gamma} - \rho_0^{\pi,\gamma}\|_{L^2(\rho_0)} + \|r_n - r_0\|_{\mathcal{H},r_0}\right\}\left\|\mathcal{T}_{k_0,\pi,\gamma}(q_n^{\pi,\gamma} - q_0^{\pi,\gamma})\right\|_{\mathcal{H},r_0} = o_p(n^{-1/2})$.

Theorem 12 (Asymptotic linearity for Example 1a)**.** *Assume D1, C5(b), and E1–E3. Then Theorem 8 holds with influence function χ_0 given in Theorem 9.*

For Example 1b, let r_n estimate r_0, let $v_n^\star$ estimate the soft Bellman fixed point $v^\star_{r_0,k_0}$, and define $\pi_n^\star(a \mid s) \propto 1\{a \in \mathcal{A}^\star\}\exp\left\{\frac{1}{\tau^\star}\{r_n(a,s) + \gamma v_n^\star(a,s)\}\right\}$. Let $q_n^\star$ estimate $q^\star_{r_0,k_0}$, set $V_n^\star(s) := \sum_{\tilde{a}\in\mathcal{A}^\star}\pi_n^\star(\tilde{a} \mid s) q_n^\star(\tilde{a},s)$, and let $\widetilde{\rho}_n^\star$ and $\rho_n^\star$ estimate $\widetilde{\rho}^\star_{r_0,k_0}$ and $\rho^\star_{r_0,k_0}$. The resulting estimator of $\psi_0 = E_0\{V^\star_{r_0,k_0}(S)\}$ is

$$
\begin{aligned}
\psi_n := \frac{1}{n}\sum_{i=1}^n V_n^\star(S_i) + \frac{1}{n}\sum_{i=1}^n \rho_n^\star(S_i)\frac{\pi_n^\star(A_i \mid S_i)}{\pi_n(A_i \mid S_i)}\{r_n(A_i,S_i) + \gamma V_n^\star(S_i') - q_n^\star(A_i,S_i)\} \\
+ \frac{\gamma}{n\tau^\star}\sum_{i=1}^n \widetilde{\rho}_n^\star(S_i)\frac{\pi_n^\star(A_i \mid S_i)}{\pi_n(A_i \mid S_i)}\left[\tau^\star \log\left\{\sum_{a\in\mathcal{A}^\star}\exp\left(\tfrac{1}{\tau^\star}\{r_n(a,S_i') + \gamma v_n^\star(a,S_i')\}\right)\right\} - v_n^\star(A_i,S_i)\right] \\
+ \frac{1}{n}\sum_{i=1}^n \left\{\tfrac{1}{\tau^\star}\widetilde{\rho}_n^\star(S_i) + \rho_n^\star(S_i)\right\}\left\{\frac{\pi_n^\star(A_i \mid S_i)}{\pi_n(A_i \mid S_i)} - 1\right\}.
\end{aligned}
$$

The fixed point $v_n^\star$ can be estimated by soft Q-iteration (Haarnoja et al., 2017; van der Laan and Kallus, 2025b), and $q_n^\star$ by fitted Q-evaluation under $\pi_n^\star$. The ratio $\widetilde{\rho}^\star_{r_0,k_0}$ can be estimated by first estimating $\rho^\star_{r_0,k_0}(s' \mid a, s)$ and then applying (13).

(E1*) (*Nuisance rates*) The following rates hold:

(a) $\|v_n^\star - v^\star_{r_0,k_0}\|_{\mathcal{H},r_0} = o_p(n^{-1/4})$.

(b) $\|\widetilde{\rho}_n^\star - \widetilde{\rho}^\star_{r_0,k_0}\|_{\mathcal{H},r_0}\left(\|r_n - r_0\|_{\mathcal{H},r_0} + \|v_n^\star - v^\star_{r_0,k_0}\|_{\mathcal{H},r_0}\right) = o_p(n^{-1/2})$.

(c) $\left(\|\rho_n^\star - \rho^\star_{r_0,k_0}\|_{\mathcal{H},r_0} + \|r_n - r_0\|_{\mathcal{H},r_0} + \|v_n^\star - v^\star_{r_0,k_0}\|_{\mathcal{H},r_0}\right)\left\|\mathcal{T}^\star_{r_0,k_0}(q_n^\star - q^\star_{r_0,k_0})\right\|_{\mathcal{H},r_0} = o_p(n^{-1/2})$.

Theorem 13 (Asymptotic linearity for Example 1b)**.** *Assume D2, C5(b), and E1*–E3*. Then Theorem 8 holds with influence function χ_0 given in Theorem 10.*

For Example 2. Let $\pi_n := \exp\{r_n\}$, let $q_n^{\nu,\gamma}$ estimate $q^{\nu,\gamma}_{r_0-g,k_0}$, and define $V_n^{\nu,\gamma}(s) := \sum_{\tilde{a}\in\mathcal{A}}\nu(\tilde{a} \mid s)\, q_n^{\nu,\gamma}(\tilde{a},s)$ and $r_{n,\nu,\gamma,g}(a,s) := g(s) + q_n^{\nu,\gamma}(a,s) - V_n^{\nu,\gamma}(s)$. Let $q_{n,\nu}^{\pi,\gamma'}$ estimate $q^{\pi,\gamma'}_{r_0,k_0,\nu}$, set $V_{n,\nu}^{\pi,\gamma'}(s) := \sum_{\tilde{a}\in\mathcal{A}}\pi(\tilde{a} \mid s)\, q_{n,\nu}^{\pi,\gamma'}(\tilde{a},s)$, and let $\rho_n^{\pi,\gamma'}$ estimate $\rho_0^{\pi,\gamma'}$. The corresponding estimator of $\psi_0 = E_0[V^{\pi,\gamma'}_{r_0,k_0,\nu}(S)]$ is

$$\begin{aligned}
\psi_n &:= \frac{1}{n}\sum_{i=1}^{n} V^{\pi,\gamma'}_{n,\nu}(S_i) \\
&+ \frac{1}{n}\sum_{i=1}^{n} \rho_n^{\pi,\gamma'}(S_i)\,\frac{\pi(A_i \mid S_i)}{\pi_n(A_i \mid S_i)}\{r_{n,\nu,\gamma,g}(A_i,S_i) + \gamma' V^{\pi,\gamma'}_{n,\nu}(S_i') - q^{\pi,\gamma'}_{n,\nu}(A_i,S_i)\} \\
&+ \frac{1}{n}\sum_{i=1}^{n} \rho_n^{\pi,\gamma'}(S_i)\left(\frac{\pi(A_i \mid S_i)}{\pi_n(A_i \mid S_i)} - \frac{\nu(A_i \mid S_i)}{\pi_n(A_i \mid S_i)}\right)\{r_n(A_i,S_i) - g(S_i) + \gamma V_n^{\nu,\gamma}(S_i') - q_n^{\nu,\gamma}(A_i,S_i)\} \\
&+ \frac{1}{n}\sum_{i=1}^{n} \rho_n^{\pi,\gamma'}(S_i)\left(\frac{\pi(A_i \mid S_i)}{\pi_n(A_i \mid S_i)} - \frac{\nu(A_i \mid S_i)}{\pi_n(A_i \mid S_i)}\right).
\end{aligned}$$

(E1)** (*Nuisance rates*) The following rate conditions hold:

(a) $\{\|\rho_n^{\pi,\gamma'} - \rho_0^{\pi,\gamma'}\|_{L^2(\rho_0)} + \|r_n - r_0\|_{\mathcal{H},r_0}\}\,\big\|\mathcal{T}_{k_0,\pi,\gamma'}(q^{\pi,\gamma'}_{n,\nu} - q^{\pi,\gamma'}_{0,\nu})\big\|_{\mathcal{H},r_0} = o_p(n^{-1/2})$.

(b) $\|r_n - r_0\|_{\mathcal{H},r_0}\left\{\big\|\mathcal{T}_{k_0,\nu,\gamma}(q_n^{\nu,\gamma} - q_0^{\nu,\gamma})\big\|_{\mathcal{H},r_0} + \big\|\mathcal{T}_{k_0,\pi,\gamma'}(q^{\pi,\gamma'}_{n,\nu} - q^{\pi,\gamma'}_{0,\nu})\big\|_{\mathcal{H},r_0}\right\} = o_p(n^{-1/2})$.

Theorem 14 (Asymptotic linearity for Example 2)**.** *Assume D1, D4, C5(b), and E1**–E3**. Then Theorem 8 holds with influence function χ_0 given in Theorem 11.*

7 Simulation study

We evaluated the proposed estimators in a continuous-state treatment MDP with two state variables, four actions, nonlinear rewards, and Gaussian state transitions. The behaviour policy is the soft-optimal policy induced by the data-generating reward and transition kernel, and all samples consist of stationary one-step transitions. Throughout the simulation study, we treat the transition kernel and discounted occupancy ratios as known, so that Bellman equations and related population quantities can be computed exactly. The aim is therefore not to study numerical error in solving Bellman equations, but to isolate the statistical problem of reward estimation and the propagation of reward-estimation error through policy and value-function operators. This design allows us to study the three policy-value targets from Section 6 within a common environment while retaining a continuous state space and sufficient structure to compute oracle values on a fine grid. All reported Monte Carlo summaries are based on 500 replications per design point.

Examples 1a and 1b illustrate complementary finite-sample behaviour. Example 1a is the most favourable setting, and debiased IRL (D-IRL) exhibits near-nominal coverage together with a clear RMSE improvement over the plug-in estimator at moderate and large sample sizes. Example 1b is more demanding because the target policy is estimated from the recovered reward, so nuisance error propagates through the soft Bellman map into both the target and the EIF correction. Even in that setting, D-IRL reduces bias and RMSE at every sample size, although coverage remains below nominal. Table 1 summarizes the results for these two examples.

Table 1: Examples 1a and 1b. Plug-in and D-IRL are summarized by Monte Carlo bias, Monte Carlo standard deviation, and RMSE. Coverage refers to the nominal 95% Wald interval for D-IRL only.

Example	n	Plug-in bias	Plug-in SD	Plug-in RMSE	D-IRL bias	D-IRL SD	D-IRL RMSE	Coverage
1a	1000	-0.661	0.235	0.702	0.851	1.025	1.331	0.980
1a	2500	-0.334	0.152	0.367	0.244	0.329	0.410	0.944
1a	5000	-0.194	0.107	0.221	0.091	0.149	0.175	0.942
1a	10000	-0.102	0.075	0.126	0.038	0.083	0.091	0.938
1b	1000	1.333	0.344	1.376	0.721	0.299	0.781	0.644
1b	2500	0.734	0.213	0.765	0.379	0.163	0.413	0.620
1b	5000	0.390	0.144	0.415	0.191	0.111	0.221	0.724
1b	10000	0.194	0.093	0.216	0.094	0.070	0.117	0.864

We report simulation results for Example 2 in Appendix A. Inference is more challenging in this setting because of the additional nuisance components, but coverage is accurate when these components are well estimated.

8 Conclusion

We developed a semiparametric framework for inference in softmax IRL and dynamic discrete choice models with nonparametric rewards. By expressing a broad class of reward-dependent estimands as smooth functionals of the behaviour policy and transition kernel, deriving their efficient influence functions, and constructing automatic debiased ML estimators, the framework delivers valid and computationally tractable inference under flexible reward representations. It therefore connects classical inference for Gumbel-shock DDC models with modern machine-learning-based IRL, and provides a reusable template for new reward-dependent targets within the shared softmax structure. Natural extensions include other shock structures, broader normalization constraints, and nonhomogeneous or finite-horizon settings.

References

P. Abbeel and A. Y. Ng. Apprenticeship learning via inverse reinforcement learning. In *ICML*, 2004.

J. H. Abbring and Ø. Daljord. Identifying the discount factor in dynamic discrete choice models. *Quantitative Economics*, 11(2):471–501, 2020.

V. Aguirregabiria and P. Mira. Dynamic discrete choice structural models: A survey. *Journal of Econometrics*, 156(1):38–67, 2010.

P. Bajari, H. Hong, and S. P. Ryan. Identification and estimation of a discrete game of complete information. *Econometrica*, 78(5):1529–1568, 2010.

R. Bellman. The theory of dynamic programming. *Bulletin of the American Mathematical Society*, 60(6):503–515, 1954.

A. Bibaut, M. Petersen, N. Vlassis, M. Dimakopoulou, and M. van der Laan. Sequential causal inference in a single world of connected units. *arXiv preprint arXiv:2101.07380*, 2021.

P. J. Bickel, C. A. Klaassen, Y. Ritov, and J. Wellner. *Efficient and adaptive estimation for semiparametric models*, volume 4. Johns Hopkins University Press Baltimore, 1993.

I. Bovenzi, A. Carmel, M. Hu, R. Hurwitz, F. McBride, L. Benac, J. R. T. Ayala, and F. Doshi-Velez. Pruning the path to optimal care: Identifying systematically suboptimal medical decision-making with inverse reinforcement learning. In *AMIA Annual Symposium Proceedings*, volume 2024, page 202, 2025.

H. Cao, S. Cohen, and L. Szpruch. Identifiability in inverse reinforcement learning. *Advances in Neural Information Processing Systems*, 34:12362–12373, 2021.

S. Chandramohan, M. Geist, F. Lefevre, and O. Pietquin. User simulation in dialogue systems using inverse reinforcement learning. In *Interspeech 2011*, pages 1025–1028, 2011.

V. Chernozhukov, D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins. Double/debiased machine learning for treatment and structural parameters, 2018.

V. Chernozhukov, W. K. Newey, and R. Singh. Automatic debiased machine learning of causal and structural effects. *Econometrica*, 90(3):967–1027, 2022.

N. Dikkala, G. Lewis, L. Mackey, and V. Syrgkanis. Minimax estimation of conditional moment models. *Advances in Neural Information Processing Systems*, 33:12248–12262, 2020.

D. Ernst, P. Geurts, and L. Wehenkel. Tree-based batch mode reinforcement learning. *Journal of Machine Learning Research*, 6, 2005.

J. Fu, K. Luo, and S. Levine. Learning robust rewards with adversarial inverse reinforcement learning. In *ICLR*, 2018.

S. Geng, H. Nassif, C. Manzanares, M. Reppen, and R. Sircar. Deep pqr: Solving inverse reinforcement learning using anchor actions. In *International Conference on Machine Learning*, pages 3431–3441. PMLR, 2020.

T. Haarnoja, H. Tang, P. Abbeel, and S. Levine. Reinforcement learning with deep energy-based policies. In *International conference on machine learning*, pages 1352–1361. PMLR, 2017.

J. Ho and S. Ermon. Generative adversarial imitation learning. In *NeurIPS*, 2016.

T. Hossain, W. Shen, A. Antar, S. Prabhudesai, S. Inoue, X. Huan, and N. Banovic. A bayesian approach for quantifying data scarcity when modeling human behavior via inverse reinforcement learning. *ACM Transactions on Computer-Human Interaction*, 30 (1):1–27, 2023.

V. J. Hotz and R. A. Miller. Conditional choice probabilities and the estimation of dynamic models. *The Review of Economic Studies*, 60(3):497–529, 1993.

V. J. Hotz, R. A. Miller, S. Sanders, and J. Smith. A simulation estimator for dynamic models of discrete choice. *The Review of Economic Studies*, 61(2):265–289, 1994.

N. Jiang and L. Li. Doubly robust off-policy value evaluation for reinforcement learning. In *International conference on machine learning*, pages 652–661. PMLR, 2016.

N. Kallus and M. Udell. Revealed preference at scale: Learning personalized preferences from assortment choices. In *Proceedings of the 2016 ACM Conference on Economics and Computation*, pages 821–837, 2016.

N. Kallus and M. Uehara. Double reinforcement learning for efficient off-policy evaluation in markov decision processes. *Journal of Machine Learning Research*, 21(167):1–63, 2020.

N. Kallus and M. Uehara. Efficiently breaking the curse of horizon in off-policy evaluation with double reinforcement learning. *Operations Research*, 70(6):3282–3302, 2022.

N. Kallus, A. M. Puli, and U. Shalit. Removing hidden confounding by experimental grounding. *Advances in Neural Information Processing Systems*, 31, 2018.

M. Kalouptsidi, P. T. Scott, and E. Souza-Rodrigues. Identification of counterfactuals in dynamic discrete choice models. *Quantitative Economics*, 12(2):351–403, 2021.

M. J. Laan and J. M. Robins. *Unified methods for censored longitudinal data and causality*. Springer, 2003.

S. Levine, Z. Popović, and V. Koltun. Nonlinear inverse reinforcement learning with gaussian processes. In *NeurIPS*, 2011.

Q. Liu, L. Li, Z. Tang, and D. Zhou. Breaking the curse of horizon: Infinite-horizon off-policy estimation. *Advances in neural information processing systems*, 31, 2018.

M. Mehrabi and S. Wager. Off-policy evaluation in markov decision processes under weak distributional overlap. *arXiv preprint arXiv:2402.08201*, 2024.

R. v. Mises. On the asymptotic distribution of differentiable statistical functions. *The annals of mathematical statistics*, 18(3):309–348, 1947.

V. Mnih, K. Kavukcuoglu, D. Silver, A. Graves, I. Antonoglou, D. Wierstra, and M. Riedmiller. Playing atari with deep reinforcement learning. *arXiv preprint arXiv:1312.5602*, 2013.

R. Munos. Error bounds for approximate value iteration. In *Proceedings of the National*

Conference on Artificial Intelligence, volume 20, page 1006. Menlo Park, CA; Cambridge, MA; London; AAAI Press; MIT Press; 1999, 2005.

R. Munos and C. Szepesvári. Finite-time bounds for fitted value iteration. *Journal of Machine Learning Research*, 9(5), 2008.

O. Nachum, Y. Chow, B. Dai, and L. Li. Dualdice: Behavior-agnostic estimation of discounted stationary distribution corrections. *Advances in neural information processing systems*, 32, 2019.

A. Y. Ng and S. J. Russell. Algorithms for inverse reinforcement learning. In *ICML*, 2000.

A. Y. Ng, D. Harada, and S. Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *Icml*, volume 99, pages 278–287. Citeseer, 1999.

A. R. Pitombeira-Neto, H. P. Santos, T. L. C. da Silva, and J. A. F. de Macedo. Trajectory modeling via random utility inverse reinforcement learning. *Information Sciences*, 660: 120128, 2024.

D. Ramachandran and E. Amir. Bayesian inverse reinforcement learning. In *IJCAI*, 2007.

M. Riedmiller. Neural fitted q iteration–first experiences with a data efficient neural reinforcement learning method. In *European conference on machine learning*, pages 317–328. Springer, 2005.

J. Rust. Optimal replacement of gmc bus engines: An empirical model of harold zurcher. *Econometrica: Journal of the Econometric Society*, pages 999–1033, 1987.

C. Shi, R. Wan, V. Chernozhukov, and R. Song. Deeply-debiased off-policy interval estimation. In *International conference on machine learning*, pages 9580–9591. PMLR, 2021.

J. Skalse and A. Abate. Partial identifiability and misspecification in inverse reinforcement learning. *arXiv preprint arXiv:2411.15951*, 2024.

J. M. V. Skalse, M. Farrugia-Roberts, S. Russell, A. Abate, and A. Gleave. Invariance in policy optimisation and partial identifiability in reward learning. In *International Conference on Machine Learning*, pages 32033–32058. PMLR, 2023.

Z. Tang, Y. Feng, L. Li, D. Zhou, and Q. Liu. Doubly robust bias reduction in infinite horizon off-policy estimation. *arXiv preprint arXiv:1910.07186*, 2019.

P. Thomas and E. Brunskill. Data-efficient off-policy policy evaluation for reinforcement learning. In *International conference on machine learning*, pages 2139–2148. PMLR, 2016.

M. Uehara, J. Huang, and N. Jiang. Minimax weight and q-function learning for off-policy evaluation. In *International Conference on Machine Learning*, pages 9659–9668. PMLR, 2020.

L. van der Laan and N. Kallus. Fitted q evaluation without bellman completeness via stationary weighting. *arXiv preprint arXiv:2512.23805*, 2025a.

L. van der Laan and N. Kallus. Stationary reweighting yields local convergence of soft fitted q-iteration. *arXiv preprint arXiv:2512.23927*, 2025b.

L. van der Laan, A. Bibaut, N. Kallus, and A. Luedtke. Automatic debiased machine learning for smooth functionals of nonparametric m-estimands. *arXiv preprint arXiv:2501.11868*, 2025a.

L. van der Laan, D. Hubbard, A. Tran, N. Kallus, and A. Bibaut. Semiparametric double reinforcement learning with applications to long-term causal inference. *arXiv preprint arXiv:2501.06926*, 2025b.

M. J. Van der Laan, S. Rose, et al. *Targeted learning: causal inference for observational* *and experimental data*, volume 4. Springer, 2011.

M. J. van Der Laan, S. Rose, M. J. van der Laan, A. Chambaz, and S. Lendle. Online targeted learning for time series. *Targeted Learning in Data Science: Causal Inference for Complex Longitudinal Studies*, pages 317–346, 2018.

J. Wang, Z. Qi, and R. K. Wong. Projected state-action balancing weights for offline reinforcement learning. *The Annals of Statistics*, 51(4):1639–1665, 2023.

M. Wulfmeier, P. Ondruska, and I. Posner. Maximum entropy deep inverse reinforcement learning. In *AAAI*, 2016.

C. Yu, J. Liu, and H. Zhao. Inverse reinforcement learning for intelligent mechanical ventilation and sedative dosing in intensive care units. *BMC medical informatics and decision making*, 19(Suppl 2):57, 2019.

R. Zhang, B. Dai, L. Li, and D. Schuurmans. Gendice: Generalized offline estimation of stationary values. *arXiv preprint arXiv:2002.09072*, 2020.

B. D. Ziebart. *Modeling Purposeful Adaptive Behavior With the Principle of Maximum Causal Entropy*. PhD thesis, Carnegie Mellon University, 2010.

B. D. Ziebart, A. L. Maas, J. A. Bagnell, A. K. Dey, et al. Maximum entropy inverse reinforcement learning. In *Aaai*, volume 8, pages 1433–1438. Chicago, IL, USA, 2008.

B. D. Ziebart, J. A. Bagnell, and A. K. Dey. Modeling interaction via the principle of maximum causal entropy. In *ICML*, 2010.

K. Zielnicki, G. Aridor, A. Bibaut, A. Tran, W. Chou, and N. Kallus. The value of personalized recommendations: Evidence from netflix. *arXiv preprint arXiv:2511.07280*, 2025.